



Saint Columban College
College of Computing Studies
Pagadian City



CanYouSee

RED 



Easy Forensics picoCTF 2025 browser_webshell_solvable

AUTHOR: SHUAILIN PAN (LECONJUROR)

Description

RED, RED, RED, RED

Download the image: [red.png](#)

Hints 

1 2 3

Red?Ged?Bed?Aed?

26,013 users solved



84% Liked



 picoCTF{FLAG}

Submit Flag

Author: The Analyst: Hyposelenia

Challenge: CanYouSee

Category: Forensics

Date: 03/27/26



I. Objective

The objective of this challenge is to analyze a compressed file and uncover hidden information within an image using **metadata analysis**.

The goal is to extract encoded data from the file's metadata and decode it to **retrieve the flag**.

II. Background

In digital forensics, important data is often hidden in places that are not immediately visible.

Three key concepts in this challenge:

- **ZIP files** → compressed containers that hold files
- **Metadata** → hidden information inside files
- **Encoding (Base64)** → a way of disguising readable data

Attackers or analysts often inspect these layers to find **hidden clues or sensitive data**.

III. Tool Used

- **Kali Linux (Oracle VirtualBox)** - Used as the working environment
- **Unzip** - Used to extract files from the archive
- **exiftool** - Used to analyze metadata
- **CyberChef** - Used to decode Base64

IV. Methodology

1. Launched **Kali Linux** in Oracle VirtualBox
2. Downloaded the file using:
wget <link_address>



```
(kali@kali)-[~/Downloads]
$ wget https://artifacts.picoctf.net/c_titan/130/unknown.zip
--2026-03-26 21:08:10-- https://artifacts.picoctf.net/c_titan/130/unknown.zip
Resolving artifacts.picoctf.net (artifacts.picoctf.net)... 18.172.21.90, 18.172.21.43, 18.172.21.26, ...
Connecting to artifacts.picoctf.net (artifacts.picoctf.net)|18.172.21.90|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2252265 (2.1M) [application/octet-stream]
Saving to: 'unknown.zip'

unknown.zip                               100%[=====]
2026-03-26 21:08:13 (1.61 MB/s) - 'unknown.zip' saved [2252265/2252265]
```

3. Extracted the ZIP file:

unzip <filename>.zip

```
(kali@kali)-[~/Downloads]
$ unzip unknown.zip
Archive: unknown.zip
  inflating: ukn_reality.jpg
```

4. Inspected the extracted file

5. Checked metadata using:

exiftool <filename>

```
(kali@kali)-[~/Downloads]
$ unzip unknown.zip
Archive: unknown.zip
  inflating: ukn_reality.jpg
(kali@kali)-[~/Downloads]
$ exiftool ukn_reality.jpg
ExifTool Version Number      : 13.25
File Name                    : ukn_reality.jpg
Directory                    : .
File Size                    : 2.3 MB
File Modification Date/Time  : 2024:03:11 20:05:55-04:00
File Access Date/Time       : 2024:03:11 20:05:55-04:00
File Inode Change Date/Time  : 2026:03:26 21:08:27-04:00
File Permissions             : -rw-r--r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : inches
X Resolution                  : 72
Y Resolution                  : 72
XMP Toolkit                  : Image::ExifTool 11.88
Attribution URL              : cGljb0NURntNRTc0RDQ3QV9ISUREM05fNmE5ZjVhYzR9Cg==
Image Width                  : 4308
Image Height                  : 2875
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample               : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:0 (2 2)
Image Size                   : 4308x2875
Meganixels                   : 12.4
```



6. Found a suspicious string in the **Attribution URL** field

```
XMP Toolkit : Image::ExifTool 11.88
Attribution URL : cGljb0NURntNRTc0RDQ3QV9ISUREM05fNmE5ZjVhYzR9Cg==
Image Width : 4308
```

7. Identified it as a **Base64-encoded string**

8. Opened **CyberChef** and applied **Base64 Decode**

9. Retrieved the **flag**

V. Result

picoCTF{ME74D47A_HIDD3N_6a9f5ac4}

Output

picoCTF{ME74D47A_HIDD3N_6a9f5ac4}



VI. Explanation

What is a ZIP file?

A ZIP file is like a **compressed folder**

Think of it like:

- Many files packed into one file
- Saves space and makes downloading easier

How do we open it?

unzip file.zip

This extracts everything inside.

What is Metadata?

Metadata is:

“Hidden information inside a file”

Example:

An image may contain:

- author name
- description
- hidden text

You can't see this just by opening the image normally.

What is exiftool?

exiftool is a tool used to:

Reveal hidden metadata inside files

Command:

exiftool image.jpg

It shows:

- camera info
- text fields
- hidden clues

What did we find?

Inside the metadata:



- A field called **Attribution URL**
 - It contained a strange string
- It didn't look normal → likely encoded

What is Base64?

Base64 is a way of **encoding data into readable text**

VII. Conclusion

This challenge highlights the importance of inspecting **file metadata during forensic analysis**.

By extracting the archive, analyzing the image's metadata, and decoding the Base64 string, the hidden flag was successfully retrieved.

This reinforces a key lesson in cybersecurity:

Not all data is visible — sometimes it is hidden in plain sight within file metadata.

— **The Analyst: Hyposelenia**